MycoDAO Project Build Report

Report Date: February 15, 2025
Project: MycoDAO
Location: `c:\Users\Abelardo\Desktop\Websites\mycodao`

Executive Summary

MycoDAO is a Next.js 14 application featuring a Bloomberg TV-style Market Pulse dashboard for tracking crypto, metals, commodities, equities, and bio assets—including the MYCO governance token. The dashboard is a **viewport-locked terminal** that fits entirely within 100vh with no vertical scrolling. This report summarizes the current build state, architecture, modules, and recent updates.

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1. Technical Stack

Technology	Version
Next.js	14.2.x
React	18.2.x
Tailwind CSS	3.4.x
Node.js	—

Port: 3004 (dev and production)
Base path: `/mycodao.financial` — app at `http://localhost:3004/mycodao.financial`

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2. Routes & Pages

Route	Purpose
`/`	Home (links to Pulse)
`/pulse`	Main Bloomberg-style dashboard (viewport-locked)
`/pulse/markets`	Expanded tickers with category filters
`/pulse/news`	News feed
`/pulse/podcasts`	Episodes list + audio player
`/pulse/learn`	Financial literacy hub
`/pulse/learn/[id]`	Lesson detail
`/pulse/myco`	Token overview
`/pulse/settings`	Refresh intervals, watchlist
`/token`	Full MYCO token page

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3. Pulse Dashboard Layout (12-column grid)

Row	Modules
1	Market Pulse Carousel (4) | Crypto (3) | Metals (2) | Big Movers (3)
2	Commodities (3) | Bio Assets (3) | Tech (3) | Business (3)
3	News (4) | Podcasts (4) | Learn (4)
4	Watchlist (3) | Market Indicators (3) | MYCO Ecosystem (3) | Research Funding (3)
5	Research (3) | Calendar/Events (3) | Quick Links (3) | Status (3)

All modules are gateways: clicking the title bar navigates to the full-page view.

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4. Viewport & Layout

- **Dashboard fits 100vh** — no vertical scroll
- **Flex column layout:** `h-screen flex flex-col overflow-hidden`
- **Fixed-height bars:** Header, BreakingNewsBand, MarketTapeStrip, BottomBar — each `h-[20px]`
- **Grid:** `gridTemplateRows: repeat(5, minmax(0, 1fr))` — rows shrink evenly
- **Grid items:** `min-h-0 overflow-hidden` for proper shrinking

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5. Design Density

- **Typography:** 9–10px
- **Title bars:** `py-[2px]`, `px-[2px]`, `text-[10px] font-semibold`
- **Rows:** `py-[1px]`, `text-[10px]`
- **Module padding:** `p-[2px]`
- **Grid gap:** `gap-[2px]`
- **Row separators:** `border-b border-neutral-800`
- **TickerRow / BigMoversRow:** `hover:bg-neutral-900` (no transitions)
- **Body:** `font-variant-numeric: tabular-nums` (globals.css)

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6. Branding

- **MycoDAO wordmark:** Top-left, tiny, muted gold (`--accent-gold` #8b7355)
- **Accents (restrained, non-crypto):**
  - **LIVE:** deep green (`--accent-green` #2d4a3e)
  - **MYCO links, selected tab:** muted gold (`--accent-gold` #8b7355)
- **No large background graphics.**

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7. Module → Full Page Mapping

Module	Destination
Crypto	`/pulse/markets?category=crypto`
Metals	`/pulse/markets?category=metals`

Commodities	`/pulse/markets?category=commodity`
Bio Assets	`/pulse/markets?category=bio`
Tech	`/pulse/markets?category=tech`
Business	`/pulse/markets?category=business`
News	`/pulse/news`
Podcasts	`/pulse/podcasts`
Learn	`/pulse/learn`
Watchlist	`/pulse/markets`
Market Indicators	`/pulse/markets?category=indicators`
MYCO Ecosystem	`/pulse/myco`
Research Funding	`/pulse/myco`
Research	`/pulse/myco`
Big Movers	`/pulse/markets`

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8. Data Architecture

Provider

- **`lib/pulse-provider.tsx`** — Fetches from API routes, provides context to all Pulse pages

Types (`lib/types.ts`)

- `Ticker` — symbol, price, changePct, sessionChangePct, sparkline, assetClass
- `NewsItem` — title, summary, source, publishedAt, category
- `PodcastEpisode` — title, show, duration, audioUrl
- `LearnModule` — title, summary, contentMd
- `MycoSnapshot` — price, changePct, researchFunding
- `ResearchFundingMetrics` — grant pool, grants deployed, proposals, votes, etc.
- `ResearchItem` — title, source, category, publishedAt

Ticker Categories

- **Crypto:** BTC, ETH, SOL, AVAX, LINK, UNI, DOT, ATOM, MYCO
- **Metals:** GOLD, SILVER, PLAT, COPPER
- **Commodities:** OIL, NATGAS, WHEAT, COPPER, CORN, SOY
- **Bio:** MYCO, BIOX, GENE, BDM
- **Tech:** AAPL, MSFT, NVDA, AMZN, GOOGL, META
- **Business:** JPM, GS, BRK.B, V, MA, UNH
- **Indicators:** DXY, SPY, VIX, US10Y

API Routes (mock-first)

- `/api/tickers` — 50+ tickers with seeded drift
- `/api/news` — 12 items

- `/api/podcasts` — 8 episodes
- `/api/learn` — 12 modules
- `/api/research` — 10 items
- `/api/myco` — MYCO snapshot + research funding metrics

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9. Components

Component	Purpose
`PulseModule`	Box module with title bar, optional href gateway
`TickerRow`	Symbol, price, change %, sparkline (grid layout, hover)
`BigMoversModule`	Top movers with All/Crypto/Trad/Bio/MYCO filters, Gainers/Losers
`BigMoversRow`	Ticker row with asset badge, session change, sparkline
`ResearchFundingModule`	Grant pool, proposals, votes, biobank metrics, timestamp
`PanelCarousel`	Rotating panels (market, news, podcast, learn, MYCO)
`BreakingNewsBand`	Top news strip
`MarketTapeStrip`	Market tape (BTC, ETH, SOL, SPY, DXY, GOLD, OIL, MYCO)
`NewsHeadline`	Compact news row
`PodcastRow`	Episode with play button
`LessonRow`	Learn module row
`ResearchRow`	Research item row
`CalendarEventsModule`	Mock CPI, Fed, earnings events
`QuickLinksModule`	Nav links
`StatusModule`	LIVE indicator, alerts
`MycoEcosystemCompact`	MYCO price, supply, token link
`BottomBar`	Time, LIVE, nav tabs

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10. Build & Run

```
npm install
npm run dev    # http://localhost:3004/mycodao.financial
npm run build
npm run start
npm run docs:pdf   # Generate PDF from update report
```

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11. Recent Updates (2025-02-15)

- **Viewport terminal:** Dashboard fits 100vh, no vertical scroll
- **Branding:** MycoDAO wordmark, restrained accents (deep green, muted gold)
- **Row separators:** `border-neutral-800` inside modules
- **Ticker hover:** `bg-neutral-900` on TickerRow, BigMoversRow
- **Typography:** `font-variant-numeric: tabular-nums` on body
- **Transitions:** Instant (no animation) on hover states

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12. Pending / TODO

- Replace mock API routes with real feeds (DEX, news API, podcast RSS, CMS)
- Connect MYCO price to live feed
- Add GitHub remote and deploy pipeline

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13. Documentation

Location	Description
`docs/update-reports/`	Update reports (this folder)
`docs/Agent-Handoff.md`	Latest work, project context
`docs/Pulse-Viewport-Terminal-2025-02-15.md`	Viewport layout
`docs/Branding-Pass-2025-02-15.md`	Branding
`docs/Pulse-Fill-Density-Polish-2025-02-15.md`	Fill + density
`docs/Pulse-*.md`	Pulse redesign and density audits

Generated for MycoDAO project build update.